CognitiveQG — Trace–Question Consistency (Case 3 & 4)

This guideline defines the trace–question consistency annotation performed on the LLaMA-2 outputs. For each argument, the model produces a reasoning trace (a five-phase analysis) followed by up to three Socratic questions. The trace and the question are evaluated **jointly** rather than in isolation, since a fluent question may rest on a flawed reasoning path and an adequate question may arise from an inadequate trace; the final question alone is therefore not diagnostic. The unit of analysis is the full **(argument, trace, question)** triple.

The 2×2 classification

Each item is classified along two axes: **trace quality**, whether the model's reasoning is aligned with the expert analysis, and **question quality**, whether the question targets the same weakness as the human reference. Crossing the two axes yields four cases:

	Question BAD	**Question GOOD**
Trace GOOD	**Case 3** — good trace, bad question ▸ ANALYZE	Case 2 — both good
Trace BAD	Case 1 — both bad	**Case 4** — bad trace, good question ▸ ANALYZE

The diagonal cases (1 and 2) are consistent by construction; these are not annotated. The off-diagonal cases (**3 and 4**) are the dissociation cases, in which the trace and the question diverge, and they constitute the annotation target. For each question in Case 3 or Case 4, the annotator determines whether it is **consistent** or **inconsistent** with the model's own preceding trace.

> *The case assignment is provided automatically from the scores (a trace is treated as good at Combined Trace Score ≥ 0.240, a question as good at GPT-Sem ≥ 0.339). Only the consistency judgment is required.*

Definitions

The annotation addresses a single question: **does the generated question target the same argumentative weakness that the model identified in its own trace?** The identified weakness is the one the model commits to in the **Evaluation** phase (its logical-fallacy diagnosis) and supports in **Explanation**; it is highlighted in the traces below.

- **Consistent** — the question targets that weakness and follows the reasoning path the trace sets out.
- **Inconsistent** — the question targets a weakness the trace did not raise, or one it explicitly dismissed; it disregards or contradicts the model's own reasoning.

> *Consistency concerns the trace–question relationship within a single generation, not agreement with the gold reference. A question may be fully consistent with its trace yet still diverge from the reference; this corresponds to Case 3.*

Example 1 — consistent (arg 49)

Argument. "I am profoundly in favor of people thinking for themselves, but only because I think it will eventually lead people to conclusions similar to mine. If you think for yourself and you arrive at a false or harmful set of views, then you wasted your time and you shouldn't have thought for yourself."

Full model reasoning trace (Phases 1–5; the diagnosed weakness is highlighted):

- **Phase 1 · Interpretation**
  - **initial understanding target:** *The argument talks about the importance of thinking for oneself.*
  - **stance target:** *against*
  - **knowledge domain:** *technical-scientific*
  - **target explanation:** *The author believes that independent thought is essential for good decision-making.*
- **Phase 2 · Analysis**
  - **core claim:** *The author believes that thinking for oneself is important.*
  - **minor claim:** *The author thinks that relying solely on others for information can lead to misinformation.*
  - **premise:** *The author presents an argument that challenges the idea of blindly following authority figures.*
  - **paraphrasing understanding:** *The author argues that independent thought is crucial for making informed decisions.*
  - **reasoning structure:** *Uses a story to illustrate the importance of independent thought.*
  - **reasoning type:** *inductive*
  - **span 1:** *The story of a person who blindly follows authority figures without questioning their actions.*
  - **span 2:** *The story of another person who questions authority and makes informed decisions based on evidence.*
  - **span 3:** *The conclusion that independent thought is necessary for good decision-making.*
  - **deductive term x:** *Sound reasoning leads to good decision-making.*
  - **deductive term y:** *Blind obedience can lead to poor decision-making.*
  - **inductive term x:** *The story illustrates the importance of independent thought.*
  - **inductive term y:** *The conclusion that independent thought is necessary for good decision-making.*
  - **has assumption:** *yes*
  - **missing component:** Evidence/logic to support the claim.
- **Phase 3 · Inference**
  - **positive consequences:** *People may make more informed decisions by thinking for themselves.*

  - **negative consequences:** *Blindly following authority figures can lead to poor decision-making.*
  - **good evidence:** *The story provides evidence for the importance of independent thought.*
  - **bad evidence:** Lack of evidence to support the claim.
  - **primary domain:** *personal*
  - **alternative type:** *applies-when*
  - **alternative keywords:** *Authority, obedience, conformity, critical thinking.*
- **Phase 4 · Evaluation**
  - **inference strength:** *2 (Moderate)*
  - **credibility factors:** *personal-bias, emotional-reasoning*
  - **logical fallacy:** hasty-generalization
  - **trustworthiness:** *can't tell (insufficient evidence)*
  - **trust explanation:** *Trustworthiness assessment is difficult due to insufficient evidence.*
- **Phase 5 · Explanation**
  - **fallacy span 1:** *No logical fallacies present.*
  - **fallacy span 2:** Hasty generalization (lack of evidence).
  - **no fallacy explanation:** *The reasoning is sound because it uses a story to illustrate the importance of independent thought.*
- **Question (CONSISTENT).** *"What evidence do you have to support the claim that independent thought leads to better decision-making?"*

✓ **Why consistent.** The trace's diagnosed weakness is a lack of supporting evidence for the central claim (Analysis, Inference, Evaluation, Explanation all converge on it). The question asks for exactly that missing evidence, so it stays on the reasoning path the trace set out → **consistent**.

Example 2 — inconsistent (arg 45)

Argument. "Because there is a difference between worshiping a god worshiping nature. I also think there's a difference between religion spirituality."

Full model reasoning trace (Phases 1–5; the diagnosed weakness is highlighted):

- **Phase 1 · Interpretation**
  - **initial understanding target:** *The argument talks about the difference between worshiping God and worshiping nature.*
  - **stance target:** *against*
  - **knowledge domain:** *technical-scientific*
  - **target explanation:** *The argument presents a clear distinction between worshiping God and nature based on biblical teachings and personal experiences.*
- **Phase 2 · Analysis**

- **core claim:** *Worshiping God versus worshiping nature.*
- **minor claim:** *There are differences between these two forms of worship.*
- **premise:** *The Bible teaches us to respect and care for creation, while nature worship implies a belief in the divine power of the natural world.*
- **paraphrasing understanding:** *Worshiping God means reverencing and obeying His commands, while worshiping nature implies recognizing and honoring the power and beauty of creation.*
- **reasoning structure:** *specific-cases*
- **reasoning type:** *inductive*
- **span 1:** *personal experience*
- **span 3:** *major-claim (recognizing the distinction between worshiping God and nature).*
- **deductive term x:** *God is the ultimate authority and creator of the universe.*
- **deductive term y:** *Nature has inherent value and beauty that should be respected and cared for.*
- **inductive term x:** *My personal experiences and observations of nature.*
- **inductive term y:** *The biblical teachings on caring for creation and recognizing the divine power of nature.*
- **missing component:** *natural beauty, power, divinity*

- **Phase 3 · Inference**
  - **positive consequences:** *Deepening our appreciation and stewardship of creation, fostering a sense of awe and wonder at the natural world.*
  - **negative consequences:** *Distracting from our relationship with the divine, leading to confusion or idolatry regarding the true nature of deity.*
  - **good evidence:** *Personal experiences, biblical teachings, scientific observations.*
  - **bad evidence:** *Lack of direct revelation from God, cultural influences, superficial interpretations of scripture.*
  - **primary domain:** *Environmental/Natural phenomena*
  - **alternative type:** *applies-when*
  - **alternative keywords:** *Nature, environment, creation, beauty, power, divinity.*
- **Phase 4 · Evaluation**
  - **inference strength:** *2*
  - **credibility factors:** *personal-bias, cultural influence, religious doctrine.*
  - **logical fallacy:** hasty-generalization
  - **trustworthiness:** *can be trusted (with caution due to the narrow scope of evidence).*
  - **trust explanation:** *Based on my understanding of biblical teachings and personal experiences, I believe this argument is moderately persuasive.*
- **Phase 5 · Explanation**
  - **fallacy span 1:** hasty generalization (from limited experiences).
  - **fallacy span 2:** false dilemma (prioritizing nature over God).

- **no fallacy explanation:** *The argument acknowledges the limitations of personal experiences and seeks to establish a broader principle through inductive reasoning.*

- **Question (INCONSISTENT).** *"What are some potential biases or heuristics that could have influenced your interpretation of the argument?"*

⚠ **Where it is inconsistent.** The trace commits to a hasty generalization from limited experiences (and a false dilemma) as the argument's weakness — see Phases 4–5, highlighted. The question does not engage that at all: it asks about the reader's own biases or heuristics, a generic self-reflection prompt unrelated to the identified fallacy. Because the question targets a different concern than the weakness the trace established, it is **inconsistent**.

Annotation procedure

- Read the trace and identify the single weakness the model settled on.
- Determine whether the question targets that weakness. If it does, label it **consistent**; if it targets a different weakness, or one the trace ruled out, label it **inconsistent**.
- When the trace is empty or records "no fallacy / none" yet the question raises a specific weakness, label the question **inconsistent**.
- Where the judgment is genuinely ambiguous, record the most defensible label and add a brief justification.

	Items	Questions	Consistent	Inconsistent
Case 3 (good trace, bad Q)	11	29	14	15
Case 4 (bad trace, good Q)	15	28	15	13
Total	26	57	29 (50.9%)	28 (49.1%)